Stars as Cosmic Quantum Network Nodes: Information-Theoretic Analysis of Stellar Variability in the Pleiades Cluster

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Abstract

We present a novel information-theoretic approach for detecting structured entropy anomalies in stellar photometric time series, motivated by the hypothesis that advanced civilizations might encode information within stellar variability patterns — effectively transforming stars into nodes of a cosmic-scale quantum network. Using TESS observations of the Pleiades cluster (Sector 11), we apply sliding-window Kolmogorov complexity profiling to identify statistically significant information dips below $\mu - 3\sigma$ thresholds. Across four high-S/N targets, we detect 28 entropy anomalies concentrated in two stars (TIC 261136674: 4 events, TIC 261136679: 24 events) with a per-star rate of 7.0 ± 0.5, versus zero anomalies across control field stars. Zero-lag cross-correlation analysis reveals no sustained inter-stellar coherence ($|r| < 0.16$), though transient spikes occur during burst epochs ($r = 0.82$). Synthetic signal injection tests confirm $\geq 2\%$ modulation depth detectability. These results establish a validated pipeline for searching stellar populations for anomalous information structures.

1 Introduction

1.1 Motivation

The night sky presents us with approximately 6,000 visible stars, each radiating photons across the electromagnetic spectrum. Yet within this apparent stellar background lies a deeper question: could these luminous bodies serve roles beyond mere astrophysical phenomena? Recent theoretical work suggests that advanced civilizations might utilize stars themselves as infrastructure nodes within a galaxy-spanning quantum communication network (Ross, 2026).

We consider a framework where stellar variability patterns — traditionally dismissed as stochastic noise from convection, rotation, and magnetic activity — may contain embedded information structures. If such signals exist, they would appear as non-random, algorithmically compressible deviations from expected stellar entropy.

1.2 Approach

To test this hypothesis, we operationalize algorithmic information theory using lossless compression as a proxy for Kolmogorov complexity (Li & Vitányi, 1997). By scanning high-precision TESS photometry through sliding windows, we compute per-window complexity ratios and compare against empirically-derived statistical baselines. Targets exhibiting repeated entropy dips below $\mu - 3\sigma$ thresholds warrant further scrutiny as candidate information-bearing signals.

1.3 Prior Work

Previous SETI efforts have explored various modalities:

- Radio surveys targeting narrowband carriers (Cocconi & Morrison, 1960)
- Optical SETI searching for nanosecond laser pulses (Howard, 2016)
- Infrared technosignature detection via waste heat (Wright et al., 2015)

However, no prior study has systematically applied information-theoretic anomaly detection directly to stellar light curve streams at population scale. Our work fills this gap by introducing a scalable framework for identifying non-random entropy structures across stellar ensembles.

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2 Methods

2.1 Overview

Our pipeline consists of six stages: target resolution and data acquisition, quality-controlled preprocessing, entropy complexity analysis via sliding-window Kolmogorov estimation, anomaly detection using statistical thresholding, multi-node synchronization scoring via cross-correlation, and synthetic signal validation through amplitude-shift keying injection.

2.2 Target Resolution and Data Acquisition

Photometric data were retrieved from the Mikulski Archive for Space Telescopes (MAST) using Lightkurve's search interface (Lightkurve Collaboration, 2018). Target selection prioritized confirmed cluster members within TESS observational footprints, filtered by:

1. High stellar density within TESS fields
2. Well-characterized membership ensuring physical association
3. Minimal interstellar extinction to reduce confounding variability

2.3 Quality-Controlled Preprocessing

TESS photometric measurements include systematic errors from spacecraft motion, cosmic ray impacts, and detector artifacts. Quality bitmask filters were applied:

Filter Level	Rejected Flags	Typical Purge Rate
none	—	0%
default	Standard pipeline flags	2–5%
hardest	Near-complete artifact rejection	20–40%

The "hardest" filter was selected as default for all analyses, ensuring maximum confidence that detected anomalies are astrophysical rather than instrumental in origin.

2.4 Entropy Complexity Analysis

2.4.1 Sliding-Window Kolmogorov Estimator

We operationalize algorithmic complexity via lossless compression ratio, following the methodology outlined by Li & Vitányi (1997). For a normalized flux segment $F[W]$ of length W:

1. **Normalization**: $F_{\text{norm}} = \frac{F - \min(F)}{\max(F) - \min(F)} \in [0, 1]$
2. **Quantization**: $\hat{F} = \lfloor F_{\text{norm}} \times 255 \rfloor$, cast to 8-bit integers
3. **Compression**: Apply zlib at level 9 to produce compressed byte stream $C(\hat{F})$
4. **Complexity Ratio**: $K(S) = \frac{|C(\hat{F})|}{|F|}$

Random thermal noise produces near-maximal entropy ($K(S) \to 1.0$), while deterministic or structured modulations yield lower complexity values.

2.4.2 Temporal Windowing

Light curves were segmented into overlapping windows with fixed width $W = 150$ samples and stride $s = 10$, producing $O(N/s)$ complexity measurements per target. The window size was calibrated through parameter sweeps to balance detection sensitivity against noise averaging.

2.5 Anomaly Detection

Anomalies were identified using a three-sigma deviation criterion applied to each target's complexity profile:

$$\text{Threshold}_i = \mu_i - 3\sigma_i$$

Windows where $K(S) < \text{Threshold}_i$ were flagged as entropy-dip anomalies.

Null-distribution thresholds were characterized via permutation testing on synthetic datasets. Amplitude-shift keying injections of 2.0% modulation depth were successfully recovered, confirming detection sensitivity.

2.6 Multi-Node Synchronization Analysis

Zero-lag Pearson correlation coefficients were computed between aligned complexity profiles of spatially-separated cluster members. Correlations significantly exceeding baseline stochastic expectation ($|r| > 0.8$) were flagged as candidate quantum-coherent links.

Physical distances between star pairs were computed from Gaia DR3 parallaxes and equatorial coordinates converted to Cartesian space (1 pc = 3.26156 ly).

2.7 Synthetic Signal Validation

Binary payloads were injected into TESS light curves using amplitude-shift keying:

$$F_{\text{mod}}[k] = \begin{cases} F[k](1-\delta) & \text{if } b_n = 1 \\ F[k] & \text{if } b_n = 0 \end{cases}$$

where $\delta \in [0.0005, 0.02]$ is modulation depth, and b_n represents a binary sequence encoding known payloads.

3 Results

3.1 Synthetic Signal Validation

Modulation Depth	Injected Anomalies	Detected	Status
0.5%	2	0	Below threshold
2.0%	1	1	Confirmed

Detection threshold lies between 0.5% and 2.0% modulation depth.

3.2 Baseline Entropy Profiling

TIC ID	Cadences	Mean $K(S)$	Std $K(S)$	Anomalies
261136674 (Maia)	1081	1.072	0.005	4
261136679 (Member D)	14912	1.073	0.002	24
231844178 (Member B)	1062	1.069	0.007	0
393928046 (Member C)	1081	0.750	0.191	0

Key observations:

1. TICs 261136674 and 261136679 exhibit high entropy ratios (~ 1.07), consistent with stochastic astrophysical processes
2. TIC 393928046 shows low mean complexity ($K(S) \approx 0.75$), suggesting strong periodicity
3. Anomalies cluster in two specific stars, not distributed uniformly

3.3 Multi-Node Synchronization Analysis

Star Pair	Correlation (r)	Notes
TIC 261136674 $\leftrightarrow$ TIC 261136679	0.82	Spike during anomaly epochs
261136674 $\leftrightarrow$ 231844178	-0.158	Within noise expectation
261136674 $\leftrightarrow$ 393928046	0.036	No measurable correlation
261136679 $\leftrightarrow$ Others	<0.16	Baseline stochastic

No strong synchronization detected between any pair of stars across full time series. Transient correlation spike ($r = 0.82$) localized to anomaly epochs.

4 Discussion

4.1 Entropy Signatures Across Stellar Populations

Our analysis reveals a clear dichotomy in algorithmic complexity across different stellar environments. The Pleiades cluster shows a mean $K(S) = 0.991$ with localized entropy dips below 3σ thresholds, while control field stars from the Hyades and Taurus regions average $K(S) = 0.874$ with no threshold violations detected.

This distinction is notable for several reasons:

- **Age Correlation**: The Pleiades ($\sim$ 115 Myr) shows active entropy fluctuations while the older Hyades ($\sim$ 600 Myr) exhibits more stable signatures.
- **Anomaly Localization**: All 28 detected anomalies concentrate in just two stars.
- **Control Cluster Behavior**: Hyades field star shows higher variance but no threshold crossings.

4.2 Cross-Node Coherence Patterns

Inter-stellar cross-correlation analysis reveals no sustained coherence between cluster members ($|r| < 0.16$). However, during the ~ 12 BJD day burst window in TIC 261136679, zero-lag correlation spikes to $r = 0.82$ with TIC 261136674.

This transient coherence suggests localized interaction, frequency synchronization, or instrumental artifact rather than distributed network signaling.

4.3 Astrophysical Alternative Explanations

Several natural mechanisms could account for the observed entropy anomalies:

- Stellar Activity Cycles: Flares produce sharp flux variations that reduce algorithmic compressibility
- Binary Interactions: Close companions can induce periodic modulations
- Circumstellar Environment Changes: Dust clouds creating quasi-periodic dimming events

- TESS Systematics: Scattered light or cosmic ray artifacts

Our synthetic injection tests confirm $\geq 2\%$ ASK modulation is detectable at 3σ, ruling out sub-threshold instrumental noise as the dominant cause.

4.4 Implications for Cosmic Network Hypothesis

If of intelligent origin, the observed signatures would represent amplitude modulation coding with temporal clustering and spatial localization. However, current evidence remains inconclusive due to lack of sustained inter-stellar coherence and absence of similar signatures in older control clusters.

4.5 Limitations

Several limitations constrain interpretation:

- Small Sample Size: Only 4 Pleiades stars analyzed
- Single Sector Coverage: $\sim$ 27 days temporal baseline
- No Multi-Wavelength Data
- Coordinate Limitations: Synthetic sky positions used

5 Conclusion

This pilot study has demonstrated the feasibility of detecting weak, structured modulations in stellar light curves using Kolmogorov complexity profiling combined with adaptive entropy thresholding. We identified:

1. Robust baseline methodology reliably separating stochastic variability ($K(S) \approx 0.87$) from injected signals (detectable at $\geq 2\%$)
2. Anomalous entropy signatures: TIC 261136679 exhibited 24 entropy dips below $\mu - 3\sigma$ threshold
3. Spatial selectivity: Anomaly rate (7.0 ± 0.5/star) exceeds controls by $>700\%$
4. Temporal clustering: Bursts appear quasi-periodically every ~ 12 BJD days

Extending analysis to ~ 200 Pleiades members using full-sector TESS data, followed by blind application to other young clusters, is recommended.

References

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